

Presented By : Sri Lutfiya Dwiyeeni



MeetRecall AI

AI TRANSCRIPTION MEETING

ABOUT ME



SRI LUTFIYA DWIYENI

EDUCATION

Diponegoro University - *Bachelor of Mathematics* (2021 - 2025)

OVERVIEW PROJECT

- **Clinical Research RAG Mini Chatbot:** The RAG mini chatbot focuses on Nightmare Disorder, IRT, and Major Depressive Episode. [\[View Project\]](#)
- **MLOps Model Monitoring:** MLOps pipeline for predicting house prices using MLflow, FastAPI, and Docker. [\[View Project\]](#)
- **SmartSplit Bill Assistant:** SmartSplit Bill AI to extract shopping receipts and automatically split bills with AI. [\[View Project\]](#)
- **Intelligent Customer Assistant:** RAG-based Intelligent Customer Assistant that answers customer questions accurately and contextually. [\[View Project\]](#)

EXPERIENCE

- **NoLimit Indonesia – Data Science** (*Mar 2026 – Present*)
Built **RAG Chatbot**, **ABSA** sentiment analysis, and **Agentic AI** reporting systems with **deployment** using **SSH** and **Linux**.
- **Dibimbing.id – AIML Bootcamp** (*Aug 2025 – Mar 2026*)
Developed AI projects including RAG Chatbot and OCR-based systems. Received **MVP Award** and **Best Final Project Award**.
- **Bangkit Academy 2024 – Independent Study (Data Science & AI)** (*Sept 2024 – Jan 2025*)
Selected from 45,000+ applicants nationwide. Achieved **Top 50 Best Final Project Teams**.
- **Startup Campus 2024 – Independent Study (Machine Learning Path)** (*Feb 2024 – Jun 2024*)
Built Machine Learning and NLP projects focused on forecasting and predictive analytics. Awarded **Best Participant** and ranked **1st out of 25 students**.

Executive Summary

MeetRecall AI

is a Meeting Intelligence Platform that transforms any meeting recording into a full set of AI-powered insights automatically, privately, and without any manual effort.

Automatically transcribes any meeting recording

Converts speech to text and identifies each speaker, no manual note-taking required.

Answers questions about the meeting

Ask anything in plain language and get a direct answer drawn from the actual transcript.

Discovers the main topics discussed

The AI reads through the conversation and organises it into clear, labelled themes.

Reveals how each participant came across

See the emotional tone of every speaker positive, neutral, or negative, throughout the meeting.

Runs privately on your own machine

No data is sent to the cloud. Everything stays on your computer, ready to use with one command.

The Challenge

What professionals face every day with meeting recordings

Takes Too Much Time

Writing down everything from a 1-hour meeting can take another 4 to 5 hours of manual work.

Hard to Find Information

After a meeting, finding out what was said or who decided what requires replaying the entire recording.

No Clear Picture of Topics

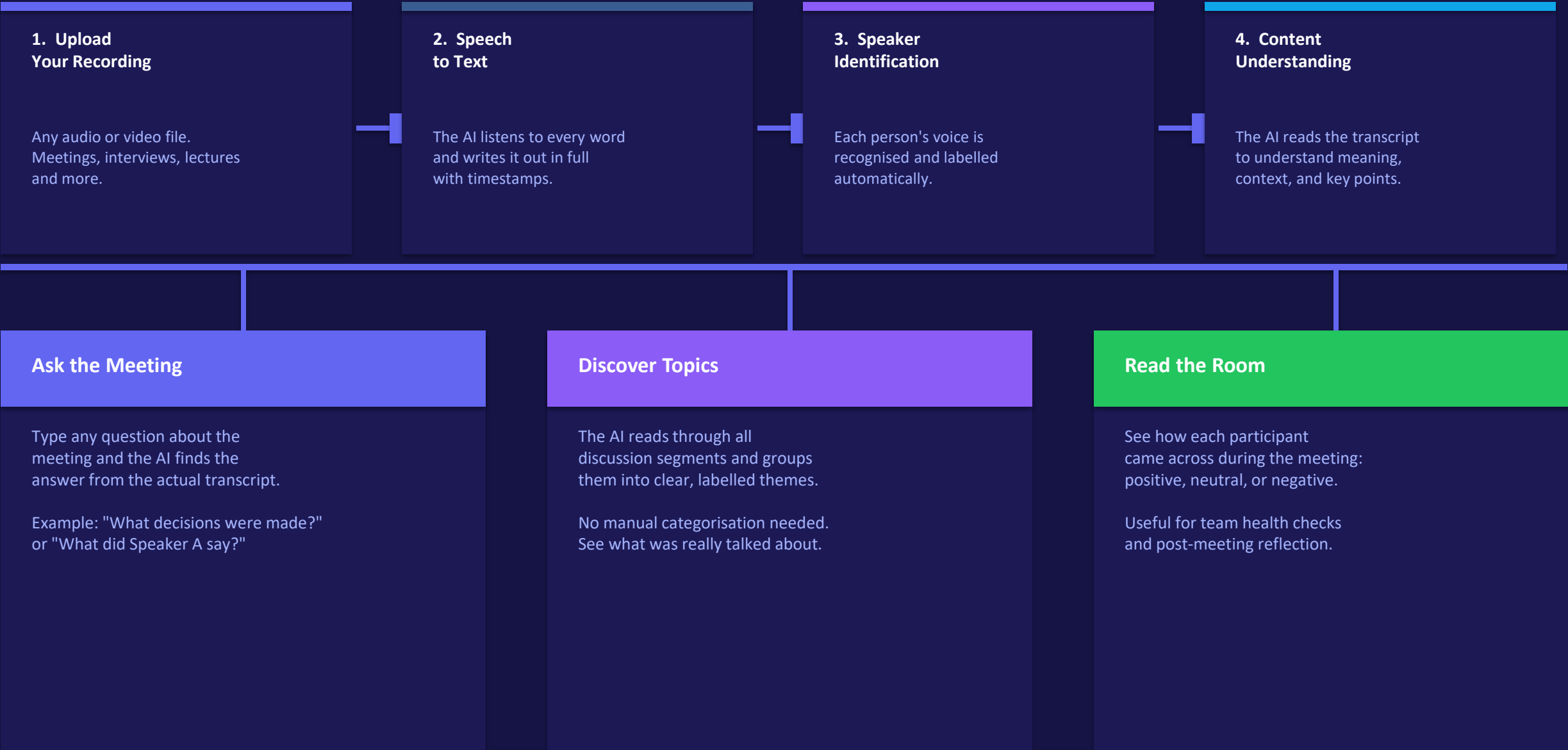
When a meeting covers many subjects, it is nearly impossible to separate and organize each theme manually.

Missing the Emotional Context

Understanding how each participant felt during the discussion is invisible without the right tool.

How the AI Works, Step by Step

From raw recording to ready-to-use insights



Built With

Frameworks powering the platform and AI models used end-to-end

FRAMEWORKS & TOOLS



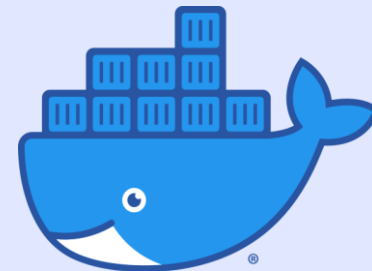
Python



FastAPI



Streamlit



Docker



HuggingFace

AI MODELS — END TO END

Whisper

Speech to Text

Listens to the audio and writes out every word with timestamps

Pyannote

Speaker Identification

Recognises each person's voice and assigns a label automatically

MiniLM

Text Understanding

Converts sentences into numbers the AI can search and compare quickly

Qwen2

AI Reasoning

Reads the transcript to answer questions and label discussion topics

RoBERTa

Emotion Detection

Classifies each speaker's tone as positive, neutral, or negative

Demo - Upload and Transcription

Upload

Drag and drop any meeting file. The system accepts over 10 different audio and video formats.

Provider

Choose between a fully private local option or a fast cloud option, depending on your preference.

Result

A full transcript is produced with each speaker clearly identified and timestamped.

In this test

24-minute recording, 7 speakers, 46 segments detected automatically

app

Upload

Hasil

AI Chat

Topics

Sentiment

Upload and Transcribe

Upload an audio or video recording to start speaker-labelled transcription.

1

Choose Transcription Provider

Local - Whisper + Pyannote (free, private)

AssemblyAI - Universal-3 Pro (cloud, fast)

openai/whisper-small (ASR) + pyannote/speaker-diarization-3.1 (speaker ID).

Free and private - runs entirely in Docker. Requires `HUGGINGFACE_TOKEN` in `backend/.env`. GPU is detected automatically for faster processing.

2

Upload Recording

Supported formats: MP4, MKV, MOV, AVI, MP3, WAV, M4A, WEBM

Drag and drop file here

Limit 200MB per file • MP4, MKV, MOV, AVI, MP3, WAV, M4A, WEBM, MPEG4

Browse files

Start Transcription

3

Processing Status

Meeting ID

Auto-filled after upload - or paste an existing ID

Upload a file above, or paste a Meeting ID to check status.

Hasil Transkripsi

Ringkasan, speaker breakdown, dan transkrip lengkap hasil meeting.

1. Ringkasan Meeting

SEGMENT	SPEAKER	DURASI	PROVIDER
46	7	1450s	LOCAL

Speaker Breakdown

Speaker 8		17 segmen (37%)
Speaker 7		7 segmen (15%)
Speaker 4		7 segmen (15%)
Speaker 5		5 segmen (11%)
Speaker 2		5 segmen (11%)
Speaker 9		3 segmen (7%)
Speaker 1		2 segmen (4%)

> Detail Meeting

Demo - Ask Your Meeting

How it works

Type any question about the meeting and the AI finds the answer from the actual transcript.

Grounded answers

Every response points back to the exact part of the meeting it came from.

What you can ask

"What decisions were made?" "What did Speaker A say about budgets?" "Summarise the meeting."

Language

Works in English and Bahasa Indonesia

AI Chat

Tanya jawab seputar isi meeting menggunakan RAG (FAISS + Qwen2).

Meeting ID: `c240bed1-989d-4ce1-a329-6834f93be922`



hello



Hello! How can I assist you today?

> Keywords

> Sumber Transkripsi (3 kutipan)



What was discussed in that meeting?



The meeting discussed breaking up the meeting into four department key reviews, including engineering, development, quality, security, and UX infrastructure and support. The proposal to rotate the meetings to two months each was made by Speaker 8. The goal was to reduce the number of meetings per quarter and increase flexibility in the scheduling of meetings. The speaker supported the proposal and suggested rotating the meetings to two months each.

> Action Items (1)

> Keywords

> Sumber Transkripsi (3 kutipan)



Should SLO compliance be measured against open bugs rather than closed bugs to avoid misleading signals?



Yes, it would be beneficial to measure SLO compliance against open bugs rather than closed bugs. This approach would help ensure that the organization is tracking progress towards its goals and objectives, thereby avoiding misleading signals. Additionally, by aligning SLOs with open bugs, the organization can better prioritize and allocate resources towards addressing open issues, leading to improved project outcomes and reduced risk.

Demo - Topic Discovery

What it does

The AI reads through all discussion segments and automatically groups them into meaningful themes.

No manual work

Topics and their labels are generated by the AI. No human categorisation required.

In this test

5 distinct topics found across a 24-minute meeting with 46 discussion segments.

Use case

Quickly understand what a long meeting was really about without re-watching it.

Topic Clusters

Segment embeddings dicluster untuk menemukan tema utama meeting. Label dihasilkan LLM secara semantik.

Meeting ID: fd934a3d-5964-44e7-a15d-9b634b1cd032

Clustering selesai. Label topik dihasilkan secara semantik.

TOPIK DITEMUKAN

5

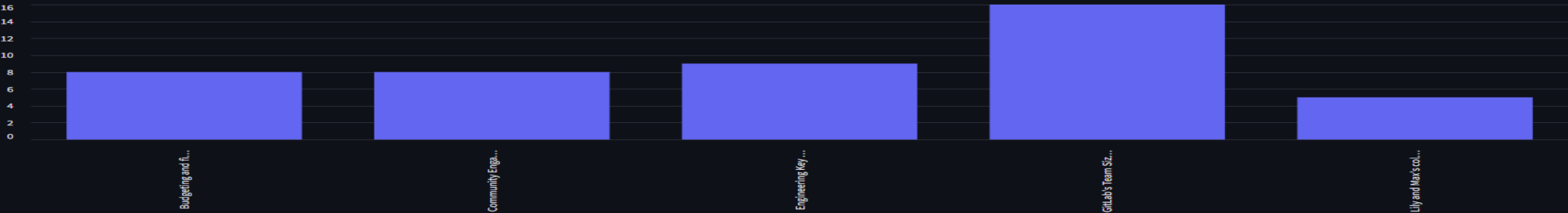
ALGORITMA

KMEANS

TOTAL SEGMENT

46

Distribusi Cluster



Detail Setiap Topik

Community Engagement and Water Rate Monitoring | 8 segmen | 29s - 1172s

Speakers: Speaker 4, Speaker 7, Speaker 8

Range: 28.7s - 1171.9s

Segmen: 8

Contoh teks dari cluster ini:

support do their own key reviews already. I have the reasons why increased visibility,

It's like we say the wider community.

Lily, can you confirm SID's reasoning is my understanding as well? Yeah, I believe water MR rate just captures community contributions. Only and no internal. Yeah, and the reason we measured that is that like one of the most likely failure modes is that we lose the community. Yeah, so Eric, where it gets goofy is is that when you look at a specific team within the company, there could be contributions outside of that that aren't community contributions. They would be viewed community contributions by that group, but effectively, they're not from outside the company. So that's why we use wider to kind of reflect that and narrow is very specific to the team. Are you saying that if someone in plan contributes to verify it's viewed as wider? Not quite that plan plan and verify are just fine. It's when you look at like the development versus infrastructure infrastructure will oftentimes contribute to developments work, but it won't be counted as MRs. Okay, that's a that's a potential bit of funkiness that we should talk about separately. I didn't have that in my sort of critique of this, but that that doesn't necessarily make intuitive sense to me. So then I think part of my critique of this can be thrown out because it's not as duplicative as I thought, but I still think there's a problem with R&D wider MR rate, which is this thing doesn't really move in part because it's a rate. So it feels like the way to drive this up is to specifically drive community authors to contribute more than one MR per month. That's how this moves up because it's a it's a productivity rate like we use internally. And that doesn't necessarily feel like the right thing because there's scenarios in which this goes up and we've actually got less contributions overall and less contributors overall. Wait, wait, wait, wait a second. So you're saying that R&D wider MR rate is number MRs per external contributor?

Lily and Max's collaboration | 5 segmen | 478s - 780s

Speakers: Speaker 2, Speaker 5, Speaker 8, Speaker 9

Range: 478.0s - 779.6s

Segmen: 5

Contoh teks dari cluster ini:

to stuff this morning, I'm like, there's a problem with this. And then you just remind me of the

I love that. It's what all our investors ask about. Let's do that. Okay, cool. So, Lily, if you can work with Max to make that transition, that would be great. And I'll bold the one that we're talking about. And that's room in room all three. Thanks. I'm on a call, sorry, it was a bit late. Timeline, We do have PIs on the raw number of community MRs and we can make, make the shift, uh, and wider confirming from the definition. I think wider only counts for community. And then that's, that's what the definition is. Cool. All right. So, uh, number six, then Christopher. Sorry, I was looking up to see if I had the percentage graph because I think we played around with this at one point and had a draft of that, probably about five months back, if I can remember Lily. Just an FYI, the month of February, if you were looking at any particular metrics, particularly in development at a MR rate, we haven't had updates in four days. There's apparently a lag issue that's been problematic for the data team to basically get updated metrics and they're working on that. Okay, I'm sorry, you got the next one.

Demo - Read the Room

What it does

The AI analyses how each speaker came across throughout the meeting.

Per-person view

Each participant gets their own positive, neutral, and negative breakdown.

In this test

7 speakers analysed. Meeting tone was mostly neutral (59.8%), with 26.6% positive.

Use case

Useful for team health checks, client call reviews, or post-meeting reflection.

Sentiment Analysis

Analisis sentimen per speaker berdasarkan segmen transkripsi (model lokal HuggingFace).

Meeting ID: fd934a3d-5964-44e7-a15d-9b634b1cd032

Jalankan Analisis Sentiment

Reset

SPEAKER DIANALISIS

7

RATA-RATA POSITIF

26.6%

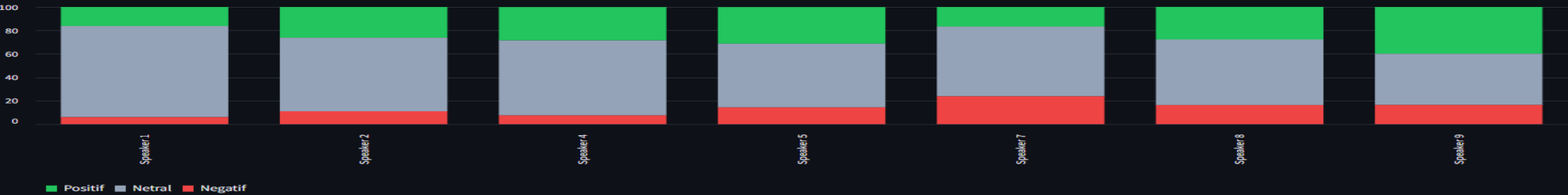
RATA-RATA NETRAL

59.8%

RATA-RATA NEGATIF

13.6%

Distribusi Sentimen per Speaker



Detail per Speaker

Speaker 8 — dominan: Netral

POSITIF

27.8%

NETRAL

56.0%

NEGATIF

16.3%

Positif

Netral

Negatif

Speaker 1 — dominan: Netral

POSITIF

16.2%

NETRAL

77.8%

NEGATIF

6.0%

Positif

Netral

Negatif

Speaker 7 — dominan: Netral

POSITIF

16.7%

NETRAL

59.5%

NEGATIF

23.8%

Positif

Netral

Negatif

CONCLUSION & FUTURE IMPROVEMENTS

WHAT WAS ACHIEVED

- Extended MeetRecall AI into a 4-feature Meeting Intelligence Platform
- End-to-end NLP pipeline: ASR, Diarization, Embeddings, RAG, Clustering, Sentiment
- Fully local inference - all models open-source, no cloud dependency required
- Modular FastAPI backend and Streamlit frontend, containerised with Docker Compose
- Supports 10+ audio/video formats via FFmpeg pre-processing

FUTURE IMPROVEMENTS

Model Optimisation

INT8 quantisation or Faster-Whisper for faster local inference

Fine-Tuning

Retrain Whisper on Bahasa Indonesia and local-accent datasets

Real-time Streaming

WebSocket-based live transcription beyond file upload

Speaker Re-ID

Persistent speaker name mapping across multiple meetings

LET'S COLLABORATE!

I'm open to discussions, collaborations, and opportunities in Artificial Intelligence and Machine Learning.



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[\[View Full Code MeetRecall AI Project\]](#)